

Functional energy and data map – electrical nets are on the child sheets.

76.2 mm envelope / eight copper layers / factory SMD assembly

Data / Control:
GPIO2/3 I2C -> U2 0x40 / U3 0x41 -> 21 PWM sinks
GPIO10/11/18 -> 3 spotlight PWM; GPIO6/7 -> CAN
GPIO0 AND ARM AND READY -> SAFE
TMP112 0x48; J18 GPIO19-23 / 3V3
Strap probes: GPIO4/5/15; USB12/13; UART16/17

Prototype note: source/current/thermal/Rf and motor-energy tests remain pending.
Open ARM = lighting inhibited. Motor stopping/holding is a separate function.

The functional energy and data map shows the following components and connections:

- J1 / external 24 V
USB: interface bias only
- F1 / Q500 / U12
Fuse, reverse block, eFuse
- V24_BUS + bulk
U501 pulse clamp
- F4-F10 -> 7 ambient zones
21 MOSFET sinks / common +24 V
- U14 -> V24_SPOT -> 3 x AL8860
U500 clamp / three paired LED outputs
- F2/F3 -> 2 CAN motors
J16 external brake / bus port
- F11 -> U10 5 V -> U11 3.3 V
ESP32-C6 / PCA9685 / CAN logic
U13 switches gate pullup bias

The schematic also includes individual component files:

- controller.kicad_sch
- power_sequence.kicad_sch
- can.kicad_sch
- usb.kicad_sch
- pwm.kicad_sch
- zone_1.kicad_sch
- zone_2.kicad_sch
- zone_3.kicad_sch
- zone_4.kicad_sch
- zone_5.kicad_sch
- zone_6.kicad_sch
- zone_7.kicad_sch
- test.kicad_sch
- power_1.kicad_sch
- power_2.kicad_sch
- spot_1.kicad_sch
- spot_2.kicad_sch
- monitor_expansion.kicad_sch
- central_led_interface.kicad_sch

ENGINEERING DRAFT – see release status before fabrication
Engineered Lighting
Sheet: /
File: engineered-lighting-rev-a.kicad_sch
Title: Integrated lighting controller
Size: A3 Date: 2026-09-08 Rev: LIGHT v0.2 ca Id: 1/20
KiCad E.D.A. 10.0.3

[illegible]

Engineered Lighting

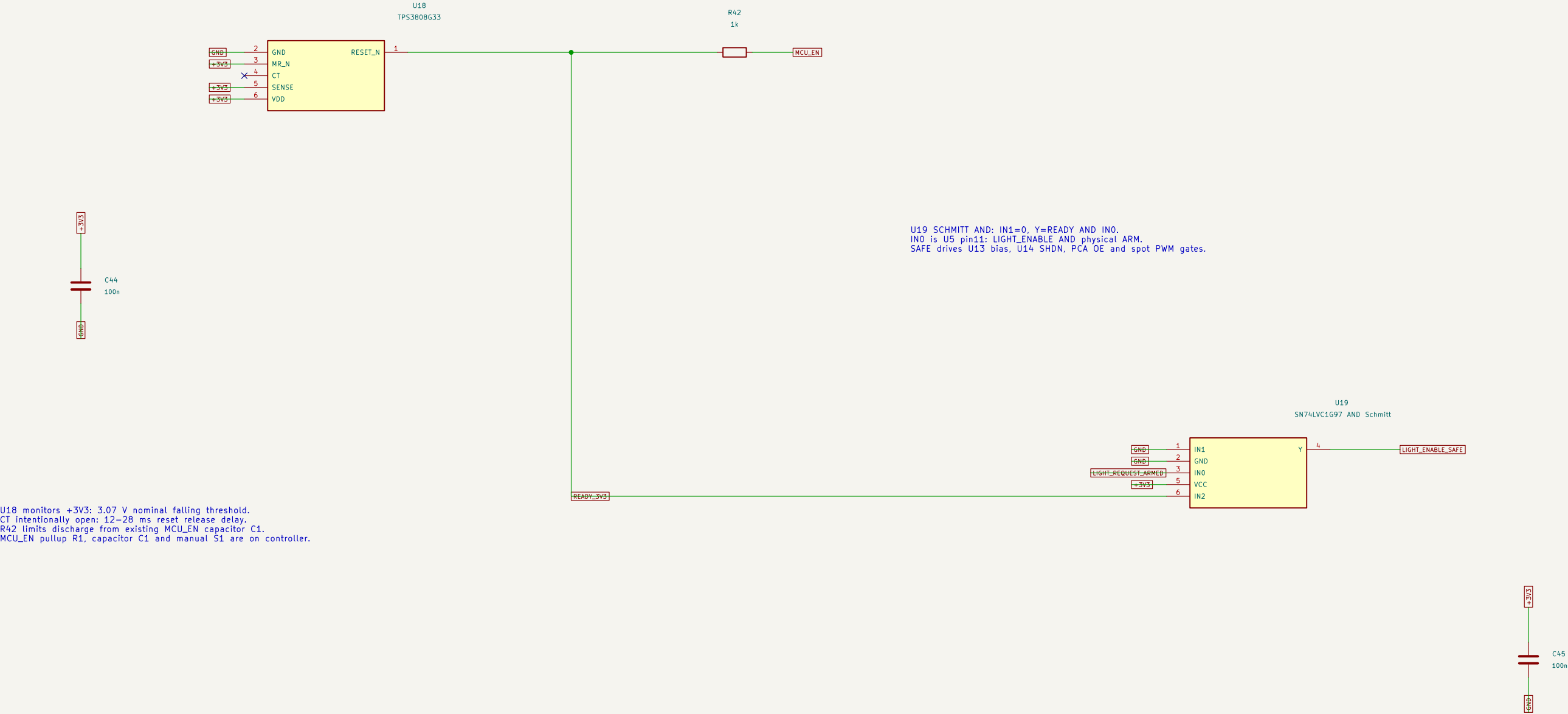
Title: Controller

Rev: A draft

Id: 2/20

POWER SEQUENCE

SUPPLY QUALIFICATION / HARDWARE LIGHTING INHIBIT



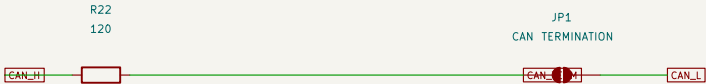
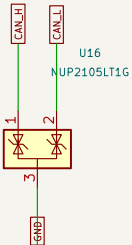
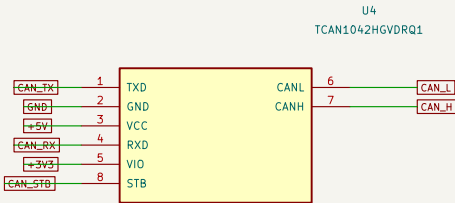
U18 monitors +3V3: 3.07 V nominal falling threshold.
CT intentionally open: 12–28 ms reset release delay.
R42 limits discharge from existing MCU_EN capacitor C1.
MCU_EN pullup R1, capacitor C1 and manual S1 are on controller.

READY also feeds USB R41/R39; independent of ARM.
Prototype timing and 0<V3V3<1.65 V behavior require measurement.
After reset: initialize outputs OFF, verify PCA state, require fresh enable.

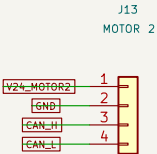
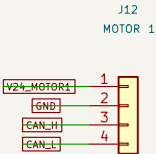
ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /power_sequence/ File: power_sequence.kicad_sch		
Title: Power Sequence		
Size: A3	Date: 2026–09–06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 3/20

CAN

CAN: 5V transceiver supply / 3.3V logic interface.
Exactly two endpoint terminations: close JP1 only if this controller is an endpoint.

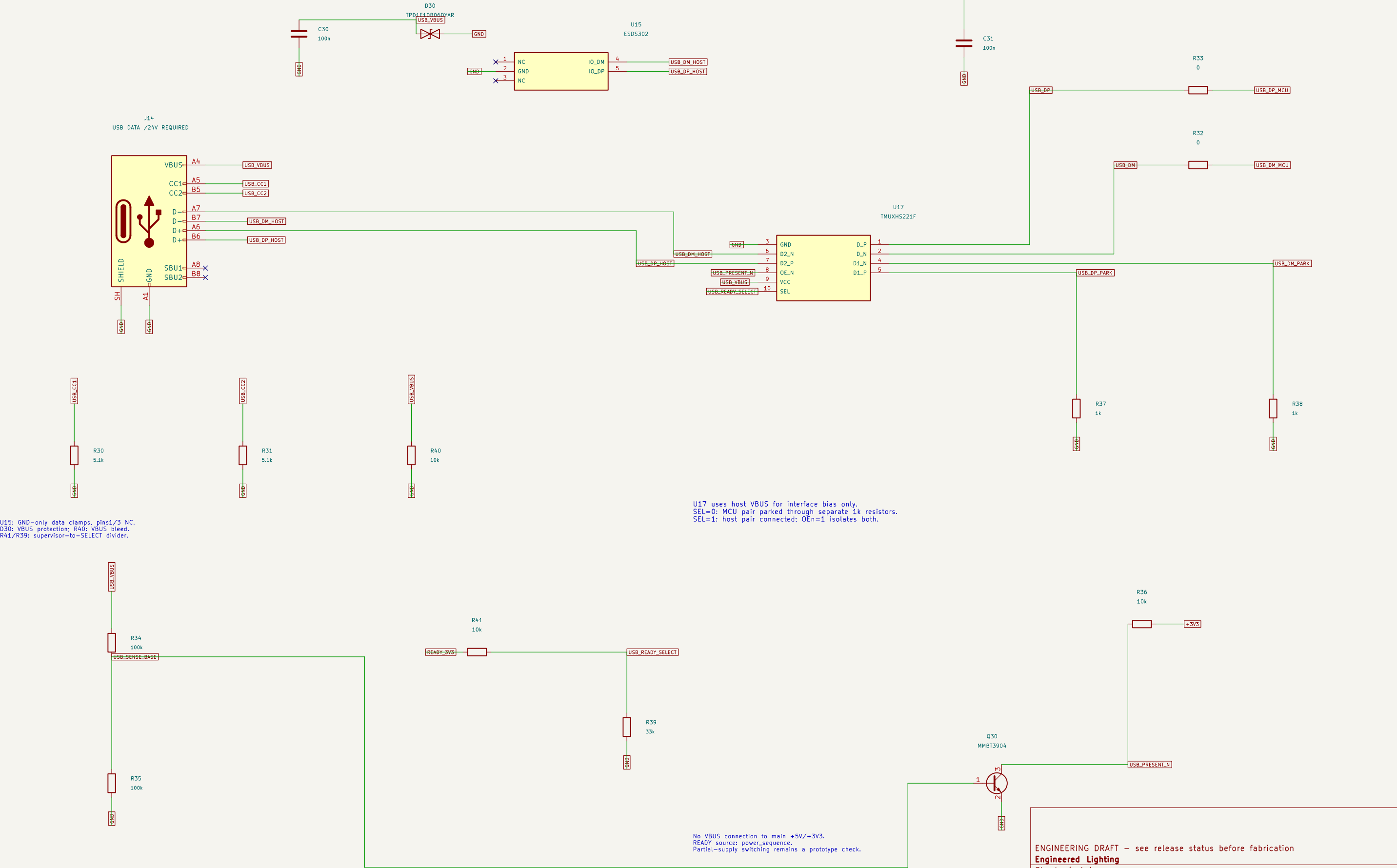


Motor connector: 1 = fused +24V, 2 = GND, 3 = CANH, 4 = CANL.
Unit-specific motor harness polarity must be verified before connection.



USB

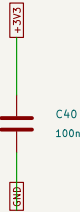
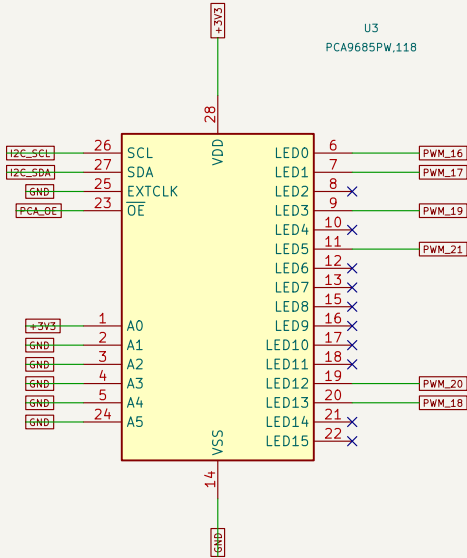
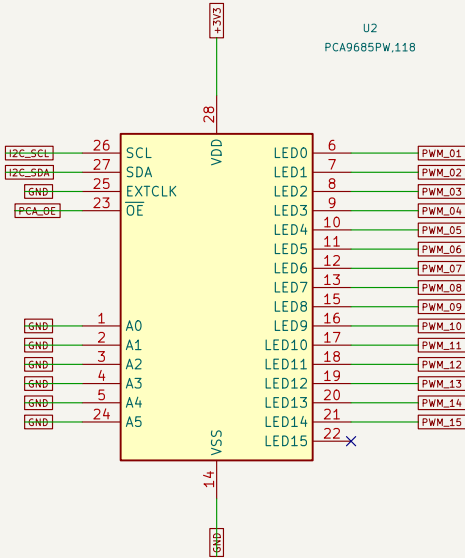
USB DATA / MAIN BOARD REQUIRES EXTERNAL 24 V



PWM

U2 address 0x40 / zones 1–5

U3 address 0x41 / zones 6–7



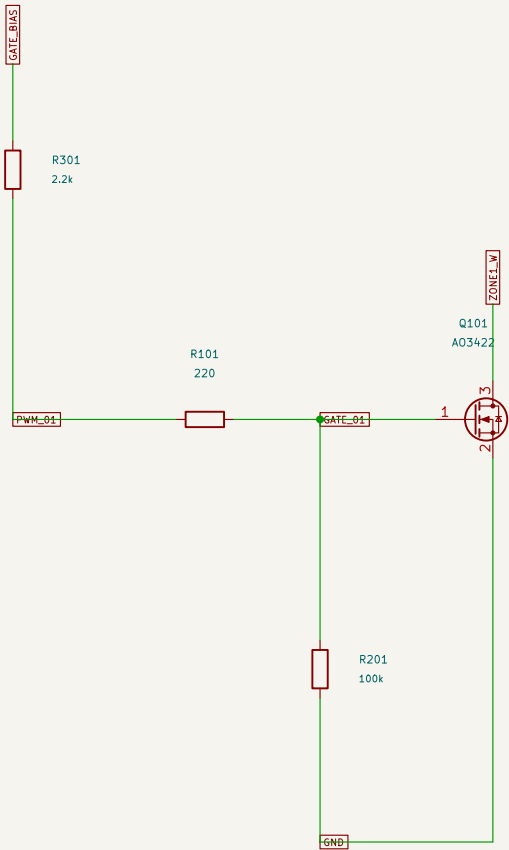
24 outputs total: 21 here + 3 direct spotlight PWM.
OUTDRV=0, OUTNE=00; INVRT/duty encoding must be verified with outputs inhibited.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /pwm/ File: pwm.kicad_sch		
Title: Pwm		
Size: A3	Date: 2026–09–06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 6/20

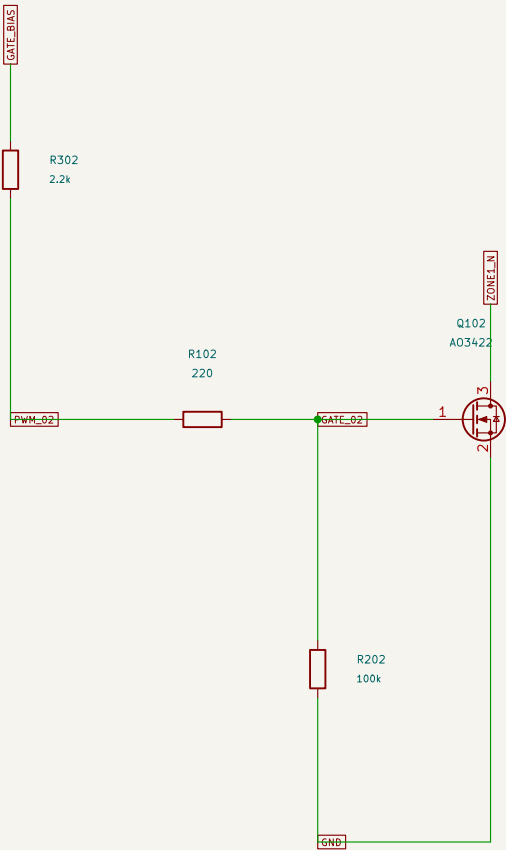
ZONE 1

24 V common positive; three independent low-side sinks.
Central J2 on centra_led_interface: +24 V pins 16/17; W 18; N 19; C 20.

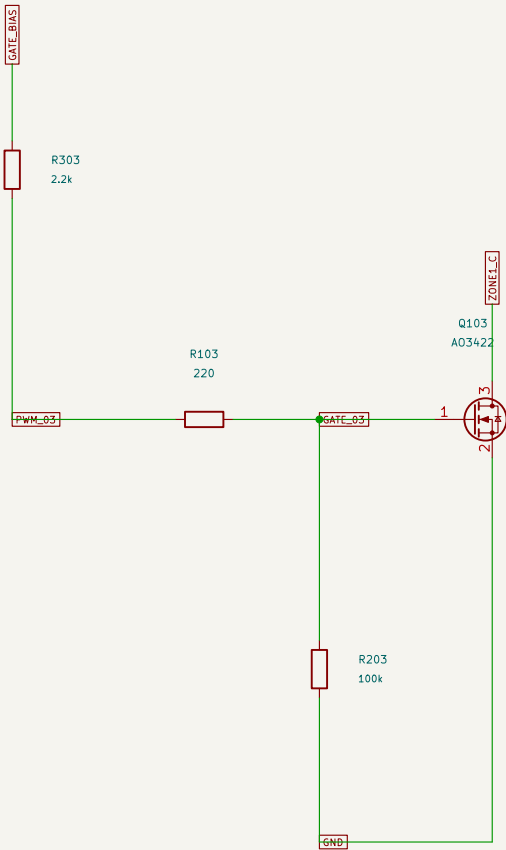
WARM / channel 1



NEUTRAL / channel 2



COOL / channel 3



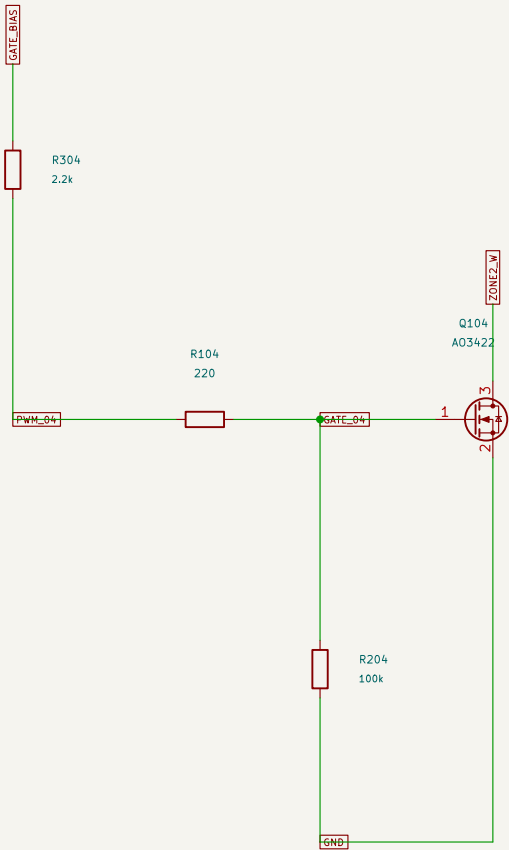
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_1/ File: zone_1.kicad_sch		
Title: Zone 1		
Size: A3	Date: 2026-09-08	Rev: LIGHT v0.2 candidate
KiCad E.D.A. 10.0.3		Id: 7/20

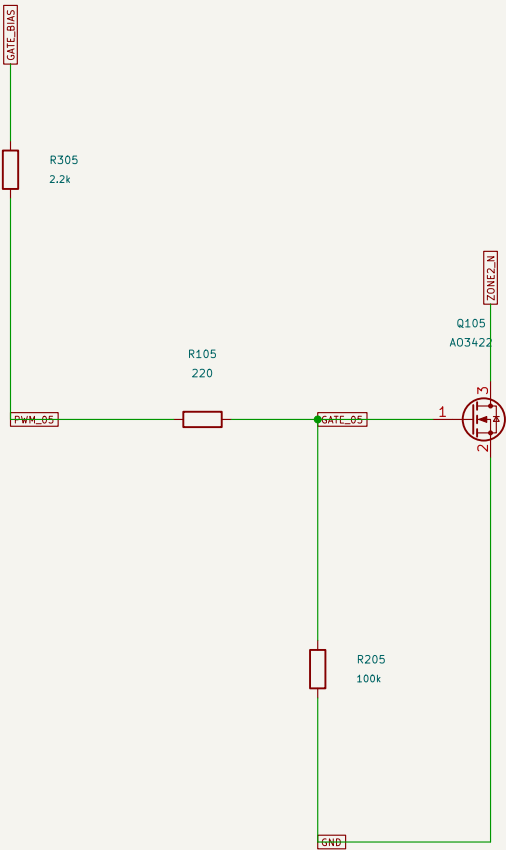
ZONE 2

24 V common positive; three independent low-side sinks.
Central J2 on centra_led_interface: +24 V pins 21/22; W 23; N 24; C 25.

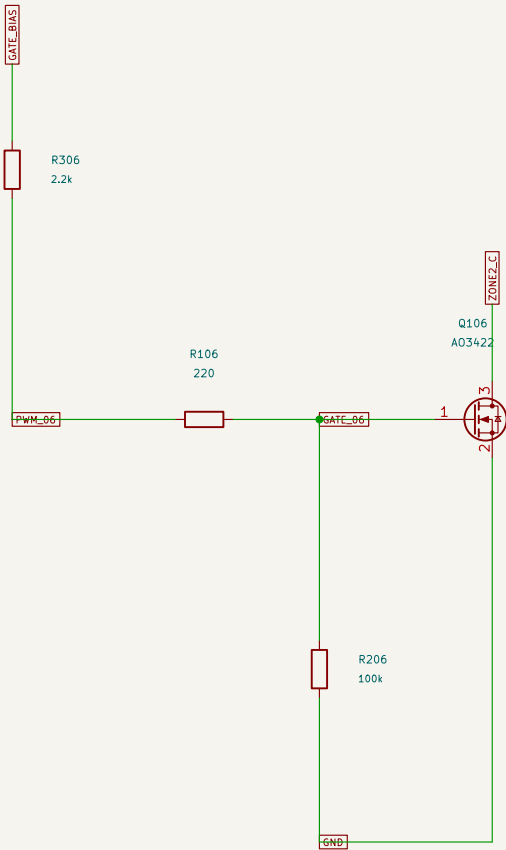
WARM / channel 4



NEUTRAL / channel 5



COOL / channel 6



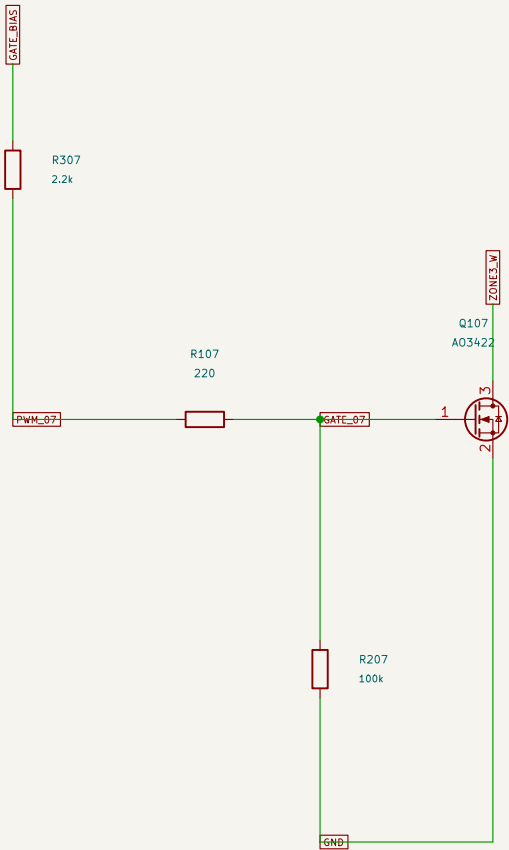
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_2/ File: zone_2.kicad_sch		
Title: Zone 2		
Size: A3	Date: 2026-09-08	Rev: LIGHT v0.2 candidate
KiCad E.D.A. 10.0.3		Id: 8/20

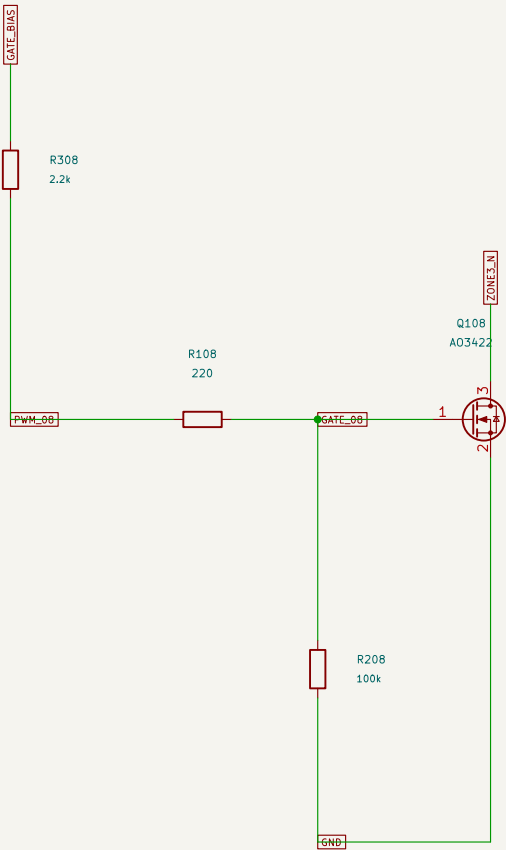
ZONE 3

24 V common positive; three independent low-side sinks.
Central J2 on centra_led_interface: +24 V pins 26/27; W 28; N 29; C 30.

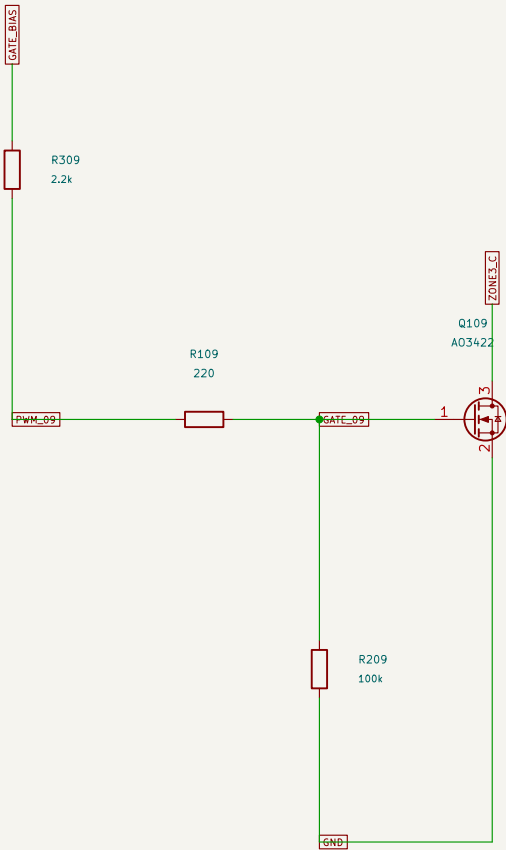
WARM / channel 7



NEUTRAL / channel 8



COOL / channel 9



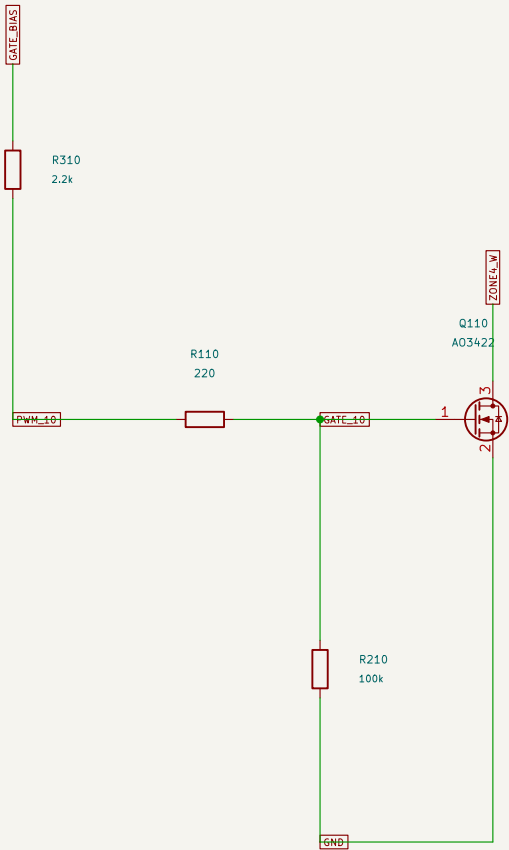
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_3/ File: zone_3.kicad_sch		
Title: Zone 3		
Size: A3	Date: 2026-09-08	Rev: LIGHT v0.2 candidate
KiCad E.D.A. 10.0.3		Id: 9/20

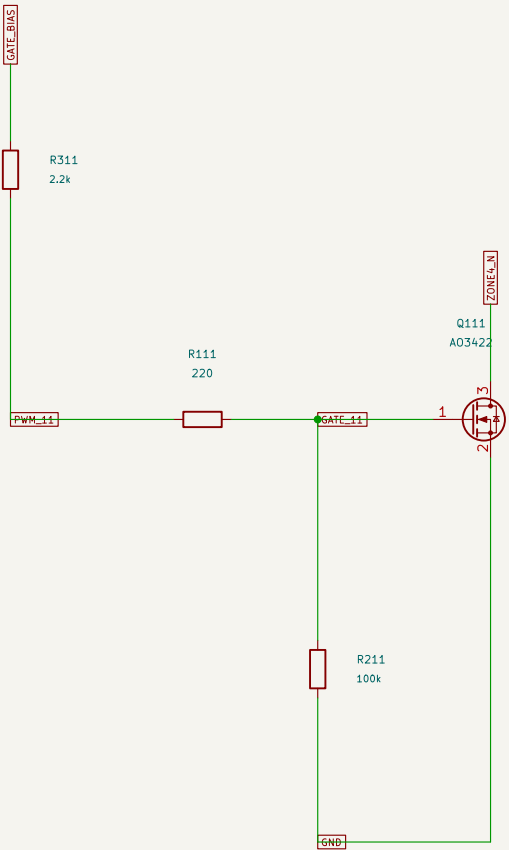
ZONE 4

24 V common positive; three independent low-side sinks.
Central J2 on centra_led_interface: +24 V pins 1/2; W 3; N 4; C 5.

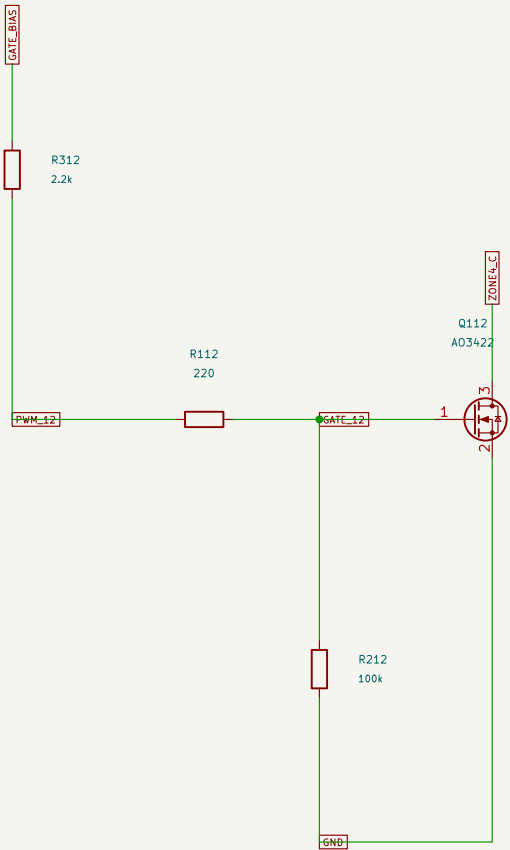
WARM / channel 10



NEUTRAL / channel 11



COOL / channel 12



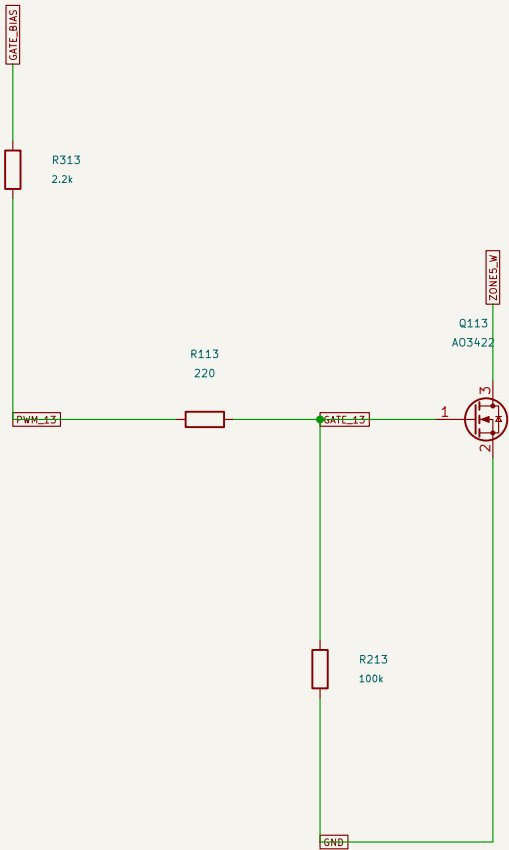
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_4/ File: zone_4.kicad_sch		
Title: Zone 4		
Size: A3	Date: 2026-09-08	Rev: LIGHT v0.2 candidate
KiCad E.D.A. 10.0.3		Id: 10/20

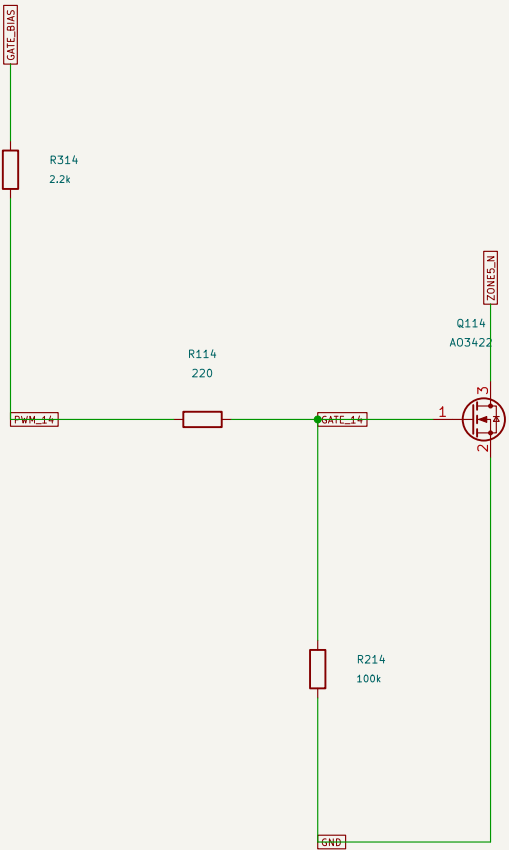
ZONE 5

24 V common positive; three independent low-side sinks.
Central J2 on centra_led_interface: +24 V pins 6/7; W 8; N 9; C 10.

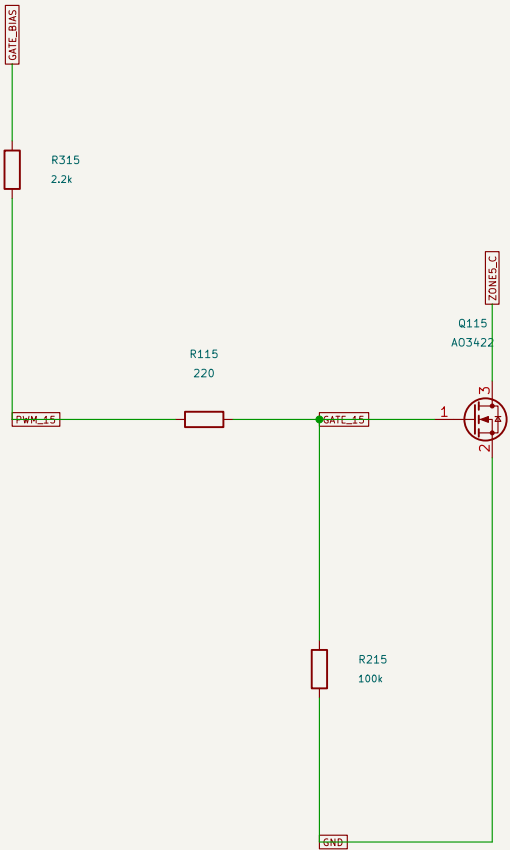
WARM / channel 13



NEUTRAL / channel 14



COOL / channel 15



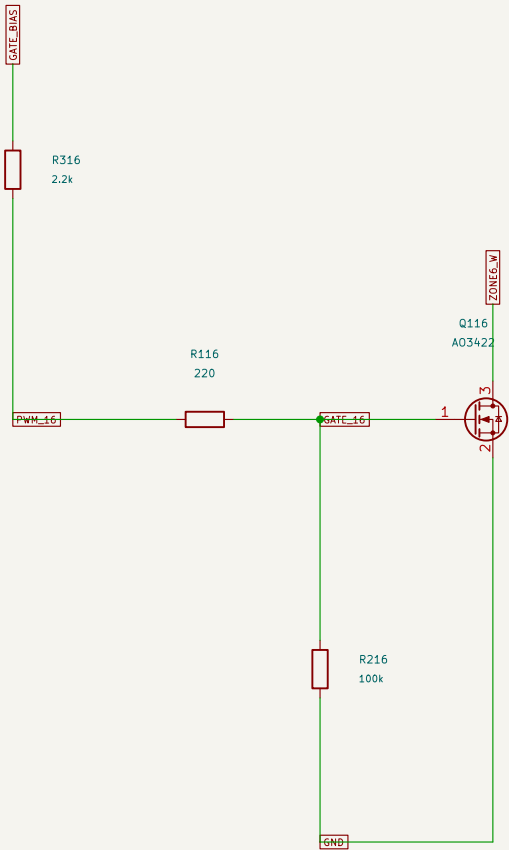
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_5/ File: zone_5.kicad_sch		
Title: Zone 5		
Size: A3	Date: 2026-09-08	Rev: LIGHT v0.2 candidate
KiCad E.D.A. 10.0.3		Id: 11/20

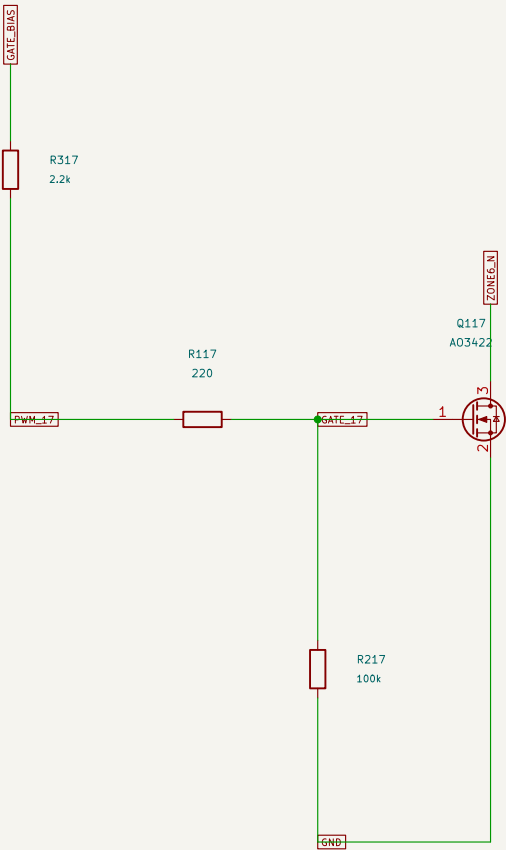
ZONE 6

24 V common positive; three independent low-side sinks.
Central J2 on centra_led_interface: +24 V pins 11/12; W 13; N 14; C 15.

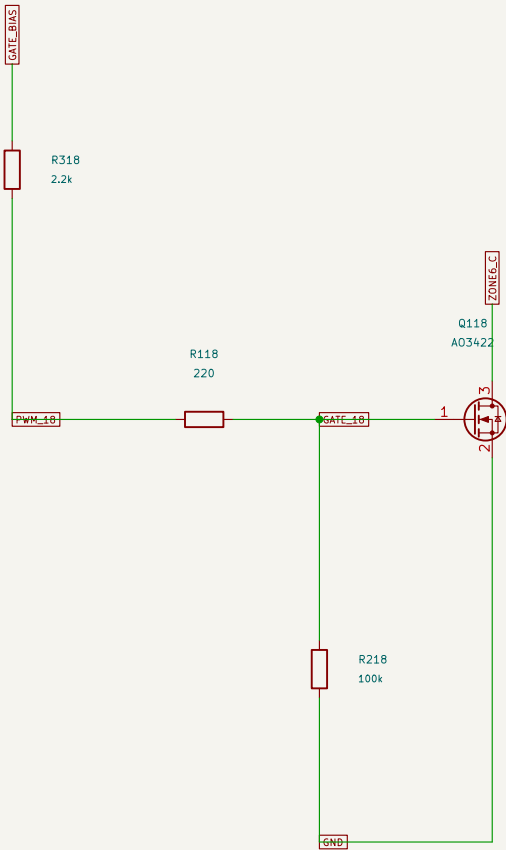
WARM / channel 16



NEUTRAL / channel 17



COOL / channel 18



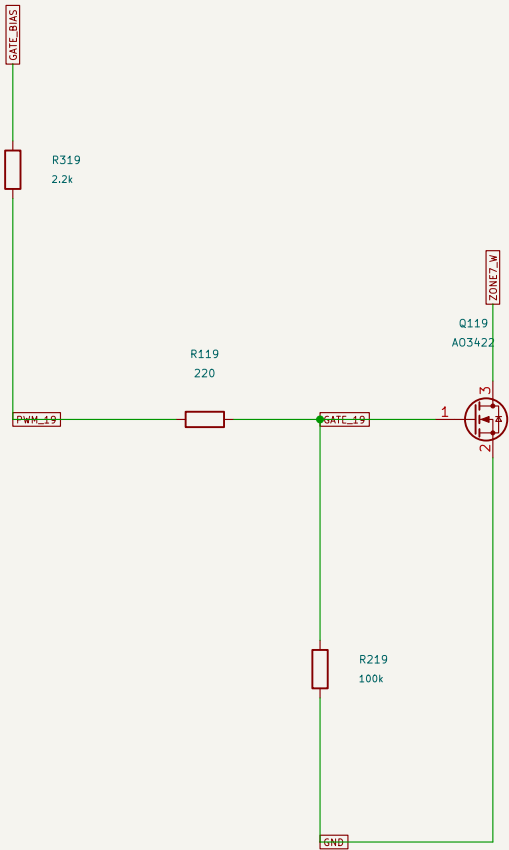
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_6/ File: zone_6.kicad_sch		
Title: Zone 6		
Size: A3	Date: 2026-09-08	Rev: LIGHT v0.2 candidate
KiCad E.D.A. 10.0.3		Id: 12/20

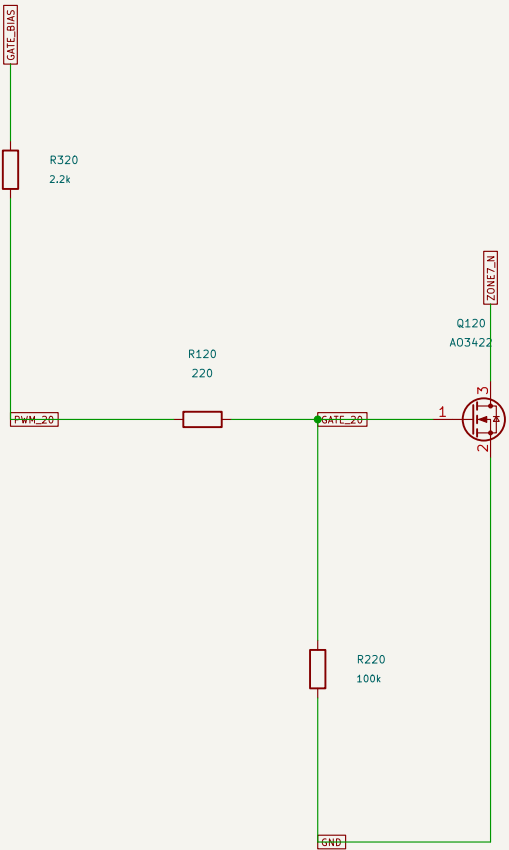
ZONE 7

24 V common positive; three independent low-side sinks.
Connector poles: 1 = fused +24 V, 2 = W, 3 = N, 4 = C.

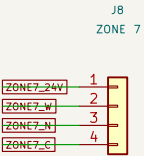
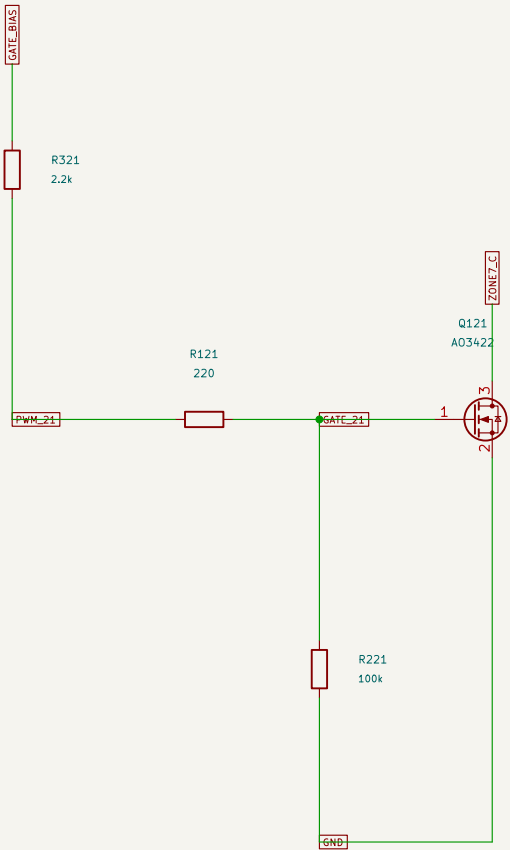
WARM / channel 19



NEUTRAL / channel 20

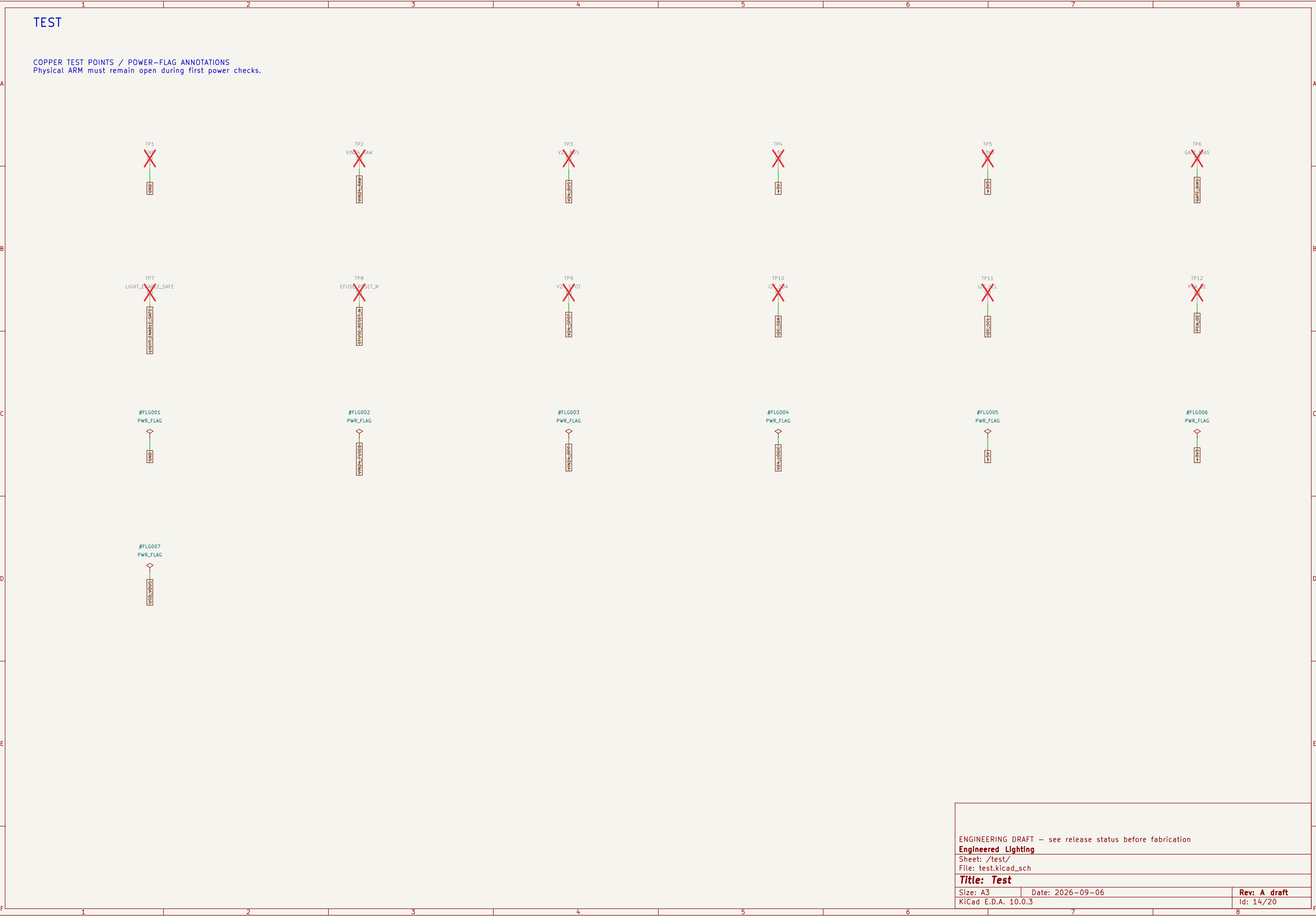


COOL / channel 21



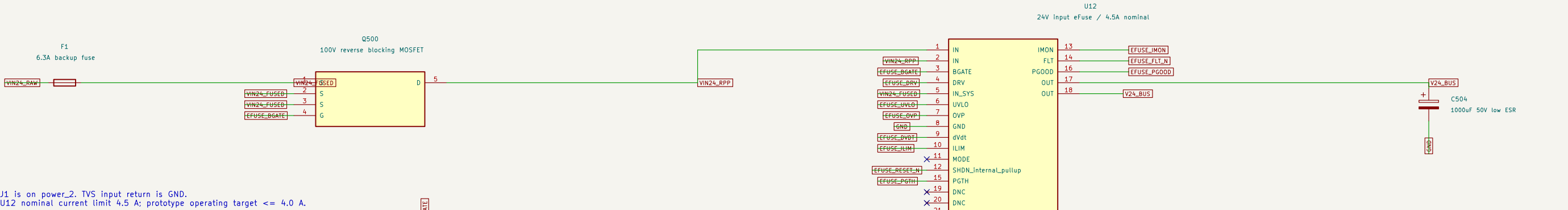
GATE_BIAS is removed by U13 when LIGHT_ENABLE_SAFE is low.
PCA9685: open-drain; disabled output LOW. Firmware duty encoding requires verification.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /zone_7/ File: zone_7.kicad_sch		
Title: Zone 7		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 13/20



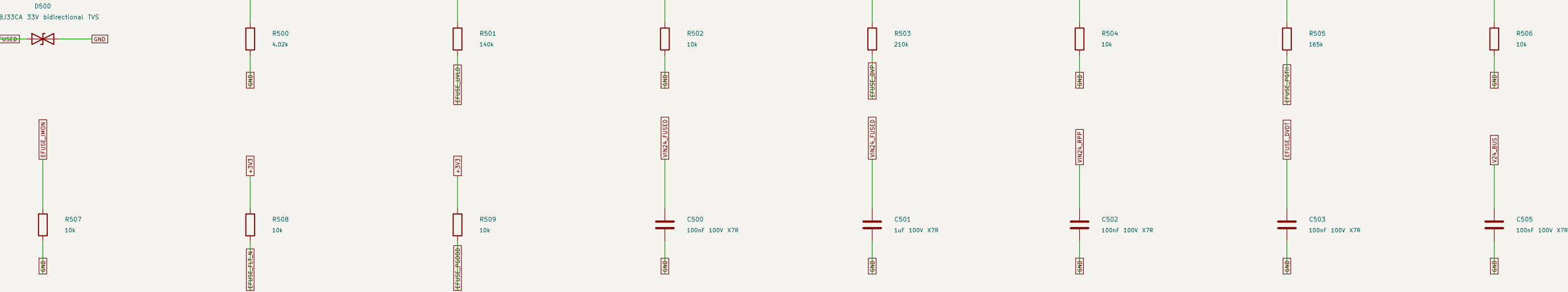
POWER 1

INPUT / REVERSE PROTECTION / MAIN eFUSE

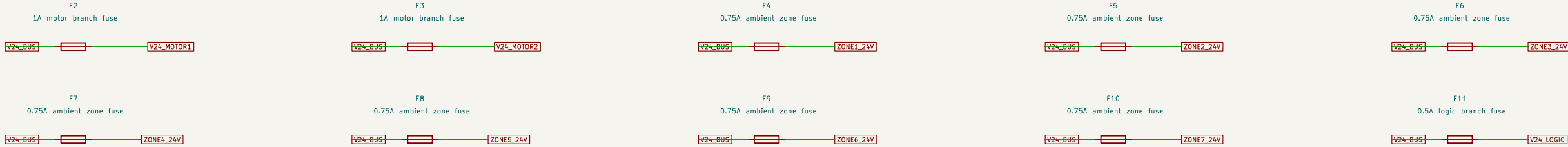


J1 is on power_2. TVS input return is GND.
U12 nominal current limit 4.5 A; prototype operating target <= 4.0 A.

Threshold / status / startup components



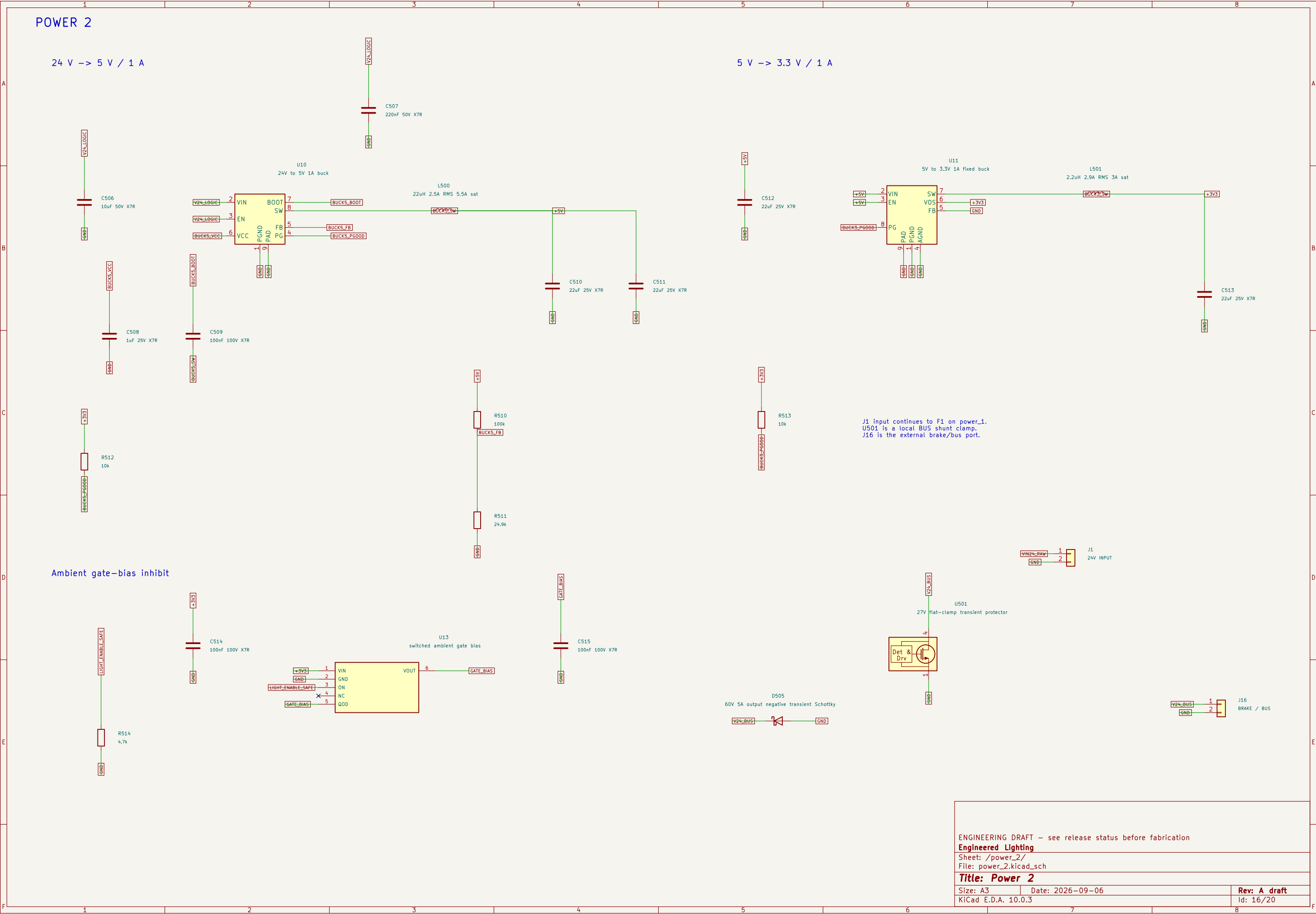
FUSED DISTRIBUTION: motors / ambient zones / logic



POWER 2

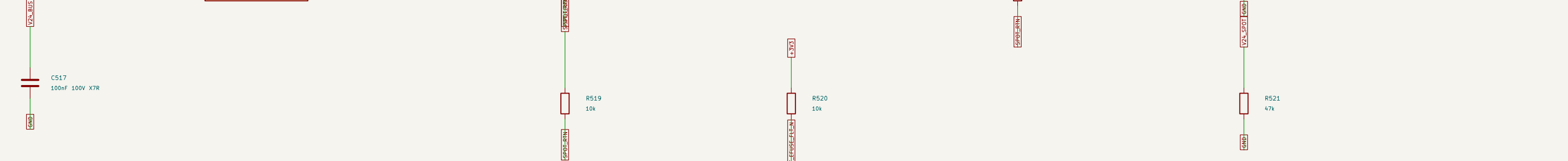
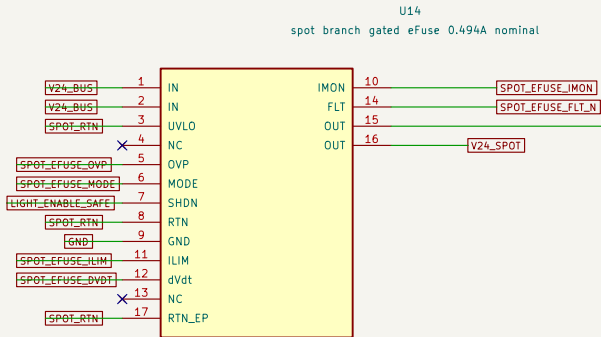
24 V -> 5 V / 1 A

5 V -> 3.3 V / 1 A



SPOT 1

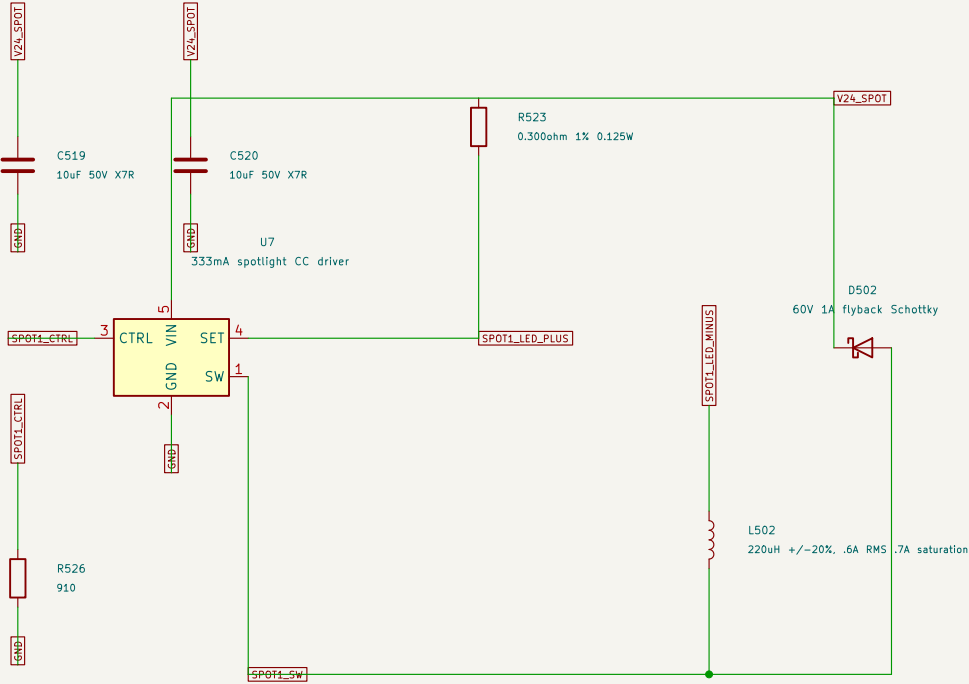
INDEPENDENT SPOT SUPPLY INHIBIT / 0.494 A NOMINAL



SPOT_RTN belongs to U14: do not join it to GND. Stored charge remains after inhibition.

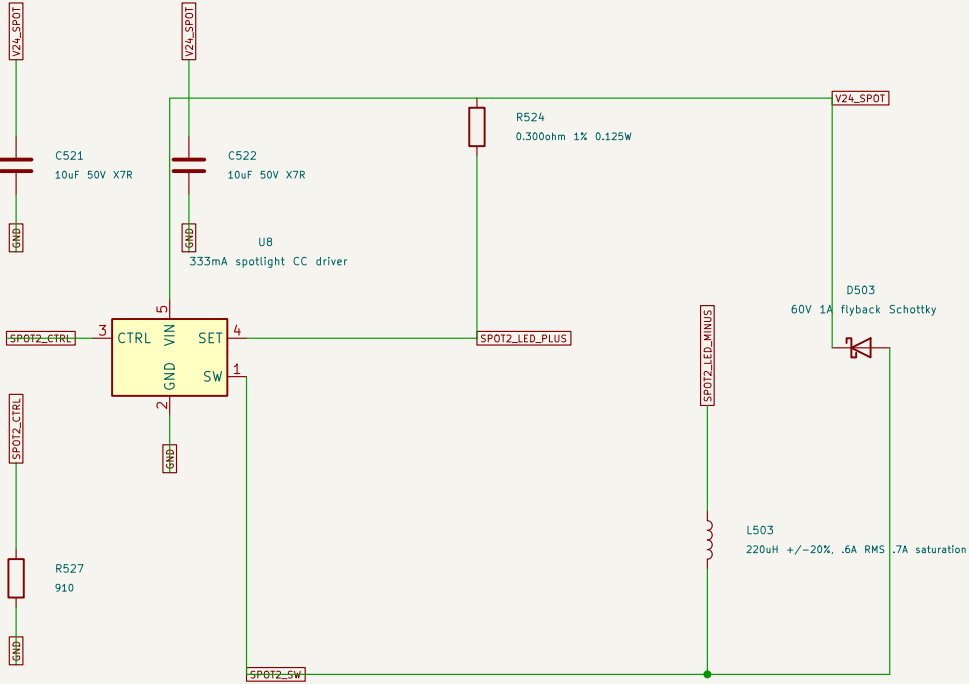
CH3 diode/input capacitors continue on spot_2.

SPOT 1 / 333 mA nominal



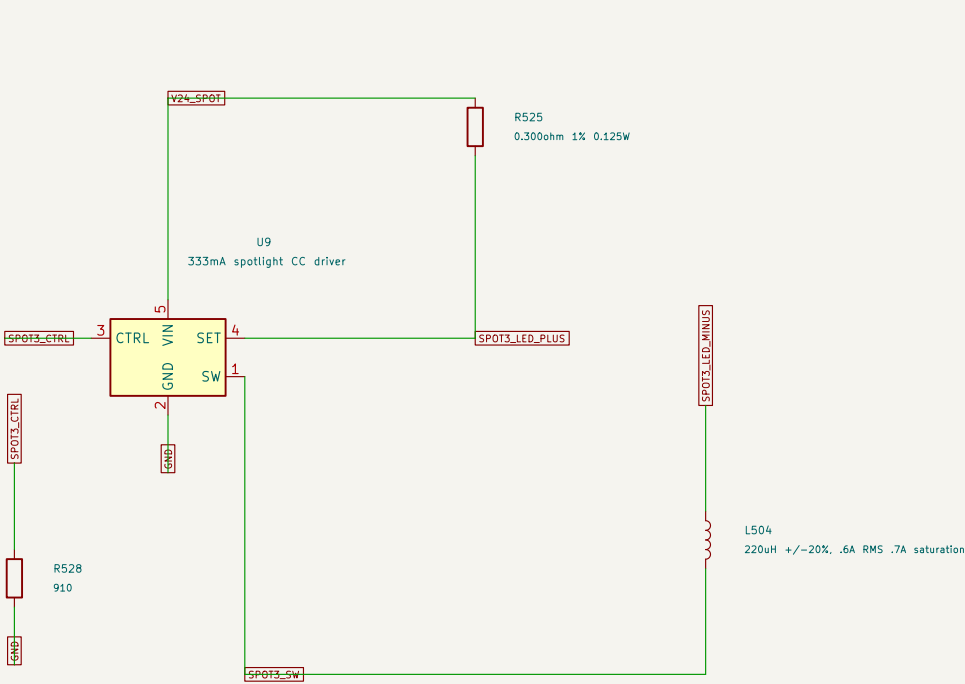
Paired LED output -> J9 / spot_2

SPOT 2 / 333 mA nominal



Paired LED output -> J10 / spot_2

SPOT 3 / 333 mA nominal



Paired LED output -> J11 / spot_2

ENGINEERING DRAFT – see release status before fabrication

Engineered Lighting

Sheet: /spot_1/

File: spot_1.kicad_sch

Title: Spot 1

Size: A3

Date: 2026-09-06

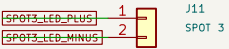
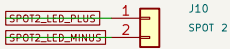
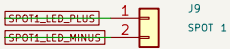
Rev: A draft

KiCad E.D.A. 10.0.3

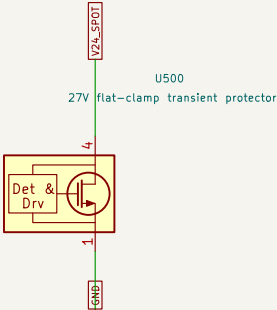
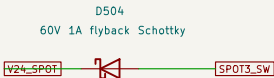
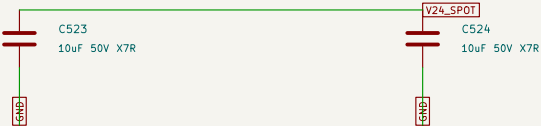
Id: 17/20

SPOT 2

PAIRED OUTPUTS: pin 1 = LED+, pin 2 = LED-. Neither return is GND.



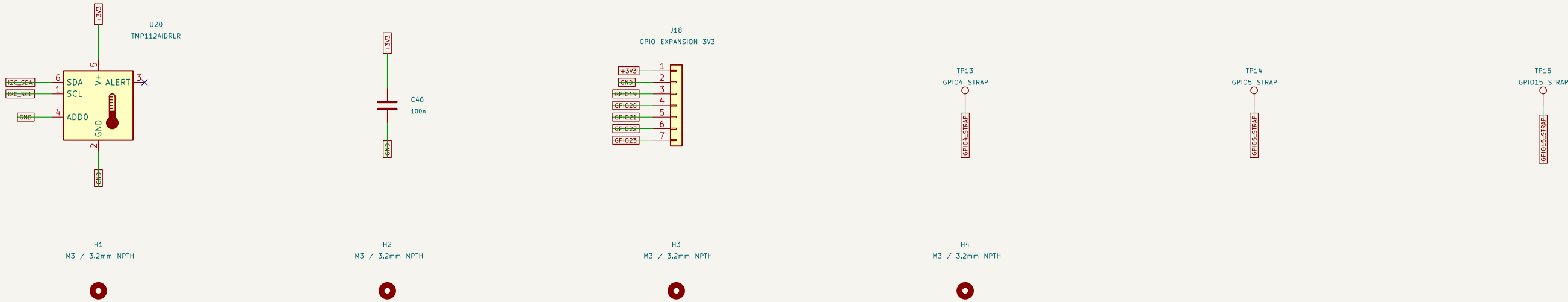
CH3 current loop continuation from U9 on spot_1



U500 local spotlight-rail shunt clamp.
Pulse limits do not establish continuous braking.

ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /spot_2/ File: spot_2.kicad_sch		
Title: Spot 2		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3	Id: 18/20	

MONITOR EXPANSION

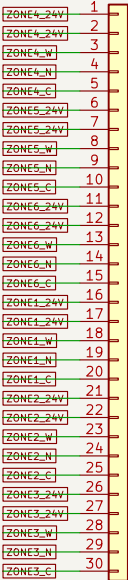


ENGINEERING DRAFT – see release status before fabrication		
Engineered Lighting		
Sheet: /monitor_expansion/ File: monitor_expansion.kicad_sch		
Title: Monitor Expansion		
Size: A3	Date: 2026-09-06	Rev: A draft
KiCad E.D.A. 10.0.3		Id: 19/20

SIX ZONES / ONE LOCKING FLEX CONNECTION

J2 mates directly to upper LED flex J100.
30 contacts carry 24 independent electrical nets.
Fused positive rails remain separate for every zone.

J2
2005280300 / 6-ZONE FLEX



PIN BLOCKS: 4 / 5 / 6 / 1 / 2 / 3
Within each block: +24 V, +24 V, W, N, C.
Duplicate positive contacts connect ONLY within that zone.
No common ground contact; these are switched LED returns.

MATING: Molex 2005280300, bottom contact / Front Flip.
Upper-flex contacts are B.Cu; reinforcement is FRONT.
Native flex front view, tip up; pin 1 LEFT.
Looking directly at gold, tip up; pin 1 RIGHT.
Gold faces main PCB when inserted.
Mate only with power disconnected; provide strain relief.

Finished insertion thickness 0.30 +/-0.05 mm, all layers included.
Fingers: gold >=0.1 um over nickel 1-5 um; no solder coating.
Tail dimensional and stack acceptance required from fabricator.
See MATING-TAIL-CONTRACT.json and fabrication drawing.

ENGINEERING DRAFT – see release status before fabrication

Engineered Lighting

Sheet: /central_led_interface/
File: central_led_interface.kicad_sch

Title: Central six-zone LED flex interface

Size: A4

Date: 2026-09-08

Rev: LIGHT v0.2 candidate

KiCad E.D.A. 10.0.3

Id: 20/20